Fiscal Year:	FY 2014	**Task Last Updated:**	FY 04/30/2014
PI Name:	Trivedi, Rohit Ph.D.		
Project Title:	Spatiotemporal Evolution of Three-Dimensional Dendritic Array Structures (SPADES)		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Project Type:	Flight	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	06/12/2012	**End Date:**	06/11/2015
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
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Flight Program:	ISS		
Flight Assignment:	ISS		
Key Personnel Changes/Previous PI:	May 2014 report: Younggil Song started as a graduate student to work on this project for his doctoral research.		
COI Name (Institution):	Karma, Alain (Northeastern University)		
Grant/Contract No.:	NNX12AK54G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The proposed research builds on insights gained through analysis of flight experimental data from the previous DSI investigation as well as associated computational simulation results, while identifying distinct research objectives for the new investigation. The proposed research involves solidification of organic alloys used as a model of metallic alloy systems. It focuses on the origin and development of the dendritic sidebranch instability that is crucial for dendrite growth and for determining the solute segregation pattern in the mushy zone that largely governs the properties of cast alloys. The primary goal of the proposed research is to quantitatively establish the fundamental physics controlling the spatiotemporal organization of the secondary sidebranch structure and its interaction with the primary array structure. Ground based experiments will be conducted to determine the appropriate sample composition and the experimental matrix of growth conditions (thermal gradient and solidification velocity) for the flight experiments. The proposed		

	research will also involve phase field simulations of the data obtained from the previous flight and and of the proposed growth conditions to aid in determination of the experimental matrix for the next set of flight experiments. These simulation results will also be used to validate the model using the flight experimental data.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Many commercial structural alloys, from nickel based superalloys used in turbine blades to light weight aluminum and magnesium alloys used for automotive parts, form dendritic microstructures during the solidification process. These microstructures in turn control the integrity and mechanical properties of cast parts. By bringing together well-controlled and accurately measured three-dimensional solidification experiments with quantitative phase-field simulations, this research should bring significant progress in understanding the origin of side-branching and the nature of the cell to dendrite transition, which have been a long standing issues in solidification for several decades. The deeper understanding of microstructure formation dynamics shall thus accelerate and optimize the design of new technological alloys that can directly benefit U.S. industries.
Task Progress:	In the cellular regime, the present microgravity experiments reveal a rich variety of 3D breathing modes with limited spatiotemporal coherence. At low growth velocities and high enough temperature gradient, cellular arrays exhibit a global oscillatory behavior with an oscillation period homogeneous over the spatially extended array. However, phase locking does not occur throughout the whole system, but only very locally among neighbor cells. Such behaviors were reproduced with phase-field simulations with a remarkable accuracy on the predicted oscillation period (e.g. experiments: t ˜ 45.6 min, phase-field: t ˜ 44.6 min, at G=28K/cm and V=1µm/s). Moreover, simulations clarified the underlying physics of this oscillatory behavior: (i) the oscillatory behavior occurs at high temperature gradients at the end of a stable spacing branch when a gap in the stable array spacing appears; (ii) the lack of global coherence in experiments is due to the spatial reorganization of cells, maintained notably by tip splitting events; (iii) breathing mode oscillations are possible over a narrow range of growth velocities for control parameters that does not lead to gap in the stable spacing branch. The identification of this subtle oscillatory mechanism constitutes the first quantitative agreement between observed and calculated dynamics in a three-dimensional configuration. In the dendritic regime, experiments on a SCN - 0.24 wt% camphor alloy suggested the existence of possible coherent modes of dendritic sidebranching for high growth velocities. This mechanism of coherent side branching was earlier reported in thin sample experiments on a dilute SCN alloy as well as in 2D and thin-sample 3D phase-field simulations, showing that this mechanism of coherent branching events is related to a limit cycle. We confirmed the existence of this coherent branching mechanism with full 3D phase-field simulations. We are currently investigating the link between the spatial order of spatially extended dendritic arrays and their dynamics of growth and sidebranching. In the DECLIC-DSI experiments, dendrites were only observed in a high velocity regime where the solidification front exhibited significant curvature due to the combination of the rejection of latent heat and the much lower thermal conductivity of the alloy than its quartz container wall. In addition, dendrite tip radii are too small to be accurately resolved at those high velocities. We expect the DSI-R experiments with a higher camphor composition (SCN - 0.5 wt% camphor) to address these limitations by reaching dendritic structures at more moderate velocities and with a higher tip radius that we could directly measure by interferometry analysis. Furthermore, uncertainties remain, particularly with respect to the accurate identification of materials and control parameters. In order to limit those uncertainties, we performed an extensive study of the SCN - 0.5 wt% camphor alloy using both ground-based thin-samples experiments, and quantitative 3D phase-field simulations. We confirmed that well-developed dendritic array structure form at a lower growth velocity, which will limit the curvature of the solidification front in future microgravity experiments. Also, measured and simulated dendrite tip radius showed very good quantitative agreement and a higher radius, which could be directly and accurately measured by interferometry. We have also undertaken a study on the critical influence of sample thickness on microstructure selection in Earth-based thin sample experiments. Simulations and ground-based experiments are in progress to develop the flight matrix of growth conditions. In addition, the implementation of thermocouples in the device would be extremely useful for reliable and direct measurement of the thermal conditions and would represent a precious information thus limiting uncertainties about control parameters in future microgravity experiments.
Bibliography Type:	Description: (Last Updated: 04/03/2024)
Articles in Peer-reviewed Journals	Bergeon N, Tourret D, Chen L, Debierre J-M, Guérin R,. Ramirez A, Billia B, Karma A, Trivedi R. "Spatiotemporal dynamics of oscillatory cellular patterns in three-dimensional directional solidification." Physical Review Letters. 2013 May 31;110(22):226102. http://dx.doi.org/10.1103/PhysRevLett.110.226102 , Jun-2013